

Hoppers Challenge: Carbon emissions and Profits

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A sustainable world is one in which our environmental systems and societies flourish, natural resources are able to replenish themselves, and people are able to live healthy and secure lives. The growing complexity and scale of scientific computing workflows and AI training pipelines place huge demands on energy and other natural resources. Sustainability can be overlooked in favour of technological advancements, therefore it is of utmost importance for us to make the effort to consider how we can fit sustainability into our own projects.

The Challenge

How can businesses grow sustainably without sacrificing profitability? Design and develop a programme that helps established businesses and emerging ventures quantify their carbon footprint, explore alternative business strategies, and identify the optimal balance between financial performance and environmental impact.

Examples

- An interactive dashboard
- A financial modelling tool that compares cost alternatives
- A carbon footprint calculator

How will we judge?

Marks will be given for the following.

- Teamwork (how much did everyone contribute)
- Originality
- Theme (How well does your project fit the theme of Sustainability?)
- Presentation Skills
- Project Quality

Extra Resources

<https://www.gov.uk/government/collections/government-conversion-factors-for-company-reporting>

<https://github.com/cloud-carbon-footprint/cloud-carbon-footprint>

<https://github.com/carbon-intensity>

Good luck!